

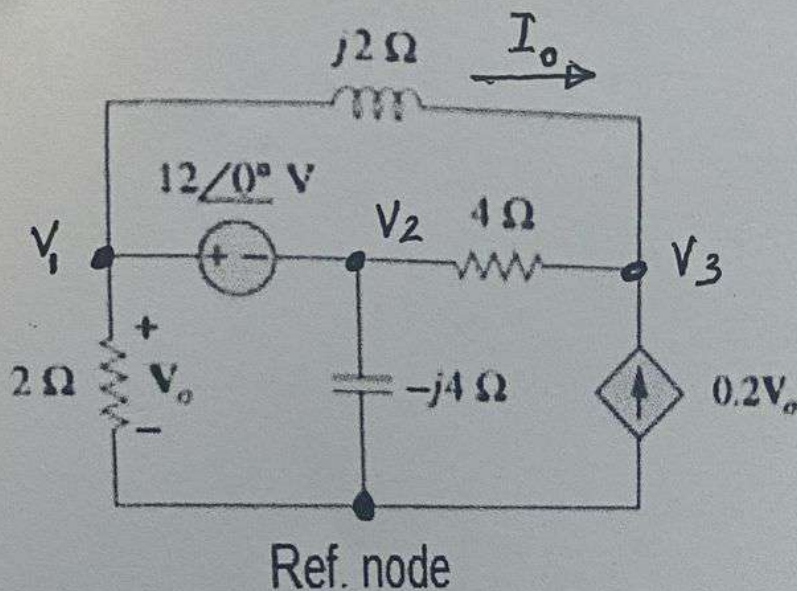


Fig.2

10) The voltage (V_0) in phasor domain is:

$V_0 = V_1 =$

11) The current (I_0) in phasor domain is:

$\frac{V_1 - V_2}{2j} =$

12) Write the required nodal equations (do not simplify):

$\frac{V_1 - V_2}{2} + \frac{V_1 - V_2}{2j} + \frac{V_2 - V_3}{4} + \frac{V_2}{-4j} = 0$ (1)

$0.2V_1 = \frac{V_3 - V_2}{4} + \frac{V_3 - V_1}{2j}$ (2)

$V_1 - V_2 = 12$ (3)

~~$\Delta = 0.1395 \angle -108.19^\circ + 0.1376 \angle -111.19^\circ$~~

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$\Delta_1 =$

$\begin{bmatrix} 0.5 - 4j & 0.25 + 0.25j \\ 0.25 + 0.25j & -0.125 + 0.125j \end{bmatrix}$
 $V_1 - V_2 = 12$

$0.125 + 0.125j, -0.125 + 0.125j$

Team
ZeroOne

Computer & Networks Engineers

prob. 1

- 1- 18° (2)
- 2- π (2)
- 3- -45.86 Volts (2)
- 4- $20 \angle -12^\circ$ A (2)
- 5- $100 \angle 6^\circ$ V (2)
- 6- 179.58 Hz (2)

10/2

2/4/1

prob 2

$$7 - 0.2 = 10 + j(2-2) = 10 \angle 0^\circ$$

$$8 - Y = \frac{1}{2} = 0.5 \text{ S}$$

$$9 - i(t) = 10 \cos(100t + 30^\circ)$$

at Node 3.

$$0.2V_1 + \frac{V_1 - V_2}{j2} = \frac{V_2 - V_3}{4}$$

Prob 5

$$10. V_0 = 7.682 \angle 50.19^\circ$$

$$11. I_0 = \frac{V_1 - V_3}{j2} = 2.16 \angle -5^\circ$$

$$V_1 - V_2 = 12 \angle 0^\circ$$

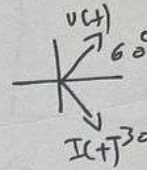
$$\frac{V_3 - V_2}{4} = \frac{V_2}{-j4} + \frac{V_1}{2} + \frac{V_1 - V_3}{j2}$$

Problem 1. For the questions 1,2,3,4,5,6 consider:

$$v(t) = 100 \cos(500t + 60^\circ) \text{ and } i(t) = 20 \sin(500t - 30^\circ).$$

1) The voltage leads the current in degrees by:

90



2) The current lags the voltage in radians by:

$\frac{90}{180} \times 2\pi = \frac{1}{2}\pi = 1.57$

3) The voltage value (v) at $t=2\text{ms}$ is:

$100 \cos(\frac{180}{\pi} \times 2 \times 10^{-3} \times 500 + 60^\circ) = -45.85$

4) The phasor domain representation of $i(t)$ is:

$1 \angle -60^\circ$

5) The phasor domain representation of $v(t)$ is:

$10 \angle -30^\circ$

6) The frequency of the voltage in Hz is:

$\frac{500}{2\pi} = 79.57$

Problem 2. For the questions 7,8,9 consider the circuit shown in figure 1 and answer:

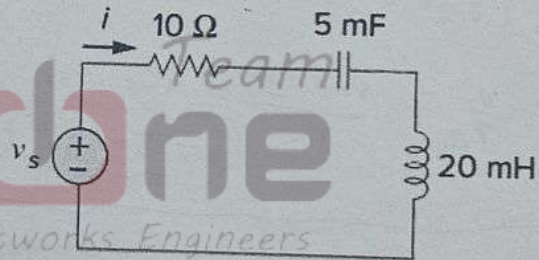


Fig. 1

7) The equivalent impedance at the terminals of the voltage source in (Ω) is:

$10 \angle 0$

8) The admittance at the terminals of the voltage source in (S) is:

$\frac{1}{10} = 0.1$

9) The steady state current (i) in time domain is:

$i(t) = 4.77 \cos(100t + 30^\circ)$

Problem 3. For the questions 10,11,12 consider the circuit shown in figure 2 and answer using nodal analysis: